



UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

SECOND SEMESTER EXAMINATION, APRIL 2026

COURSE NO : 361

COURSE NAME: PROBABILITY AND STATISTICS

CLASS: GM/CV/ES/GL III

TIME: 3 HOURS

Name: _____

Index Number: _____

SECTION A

Answer all objective questions using the provided MCQ answer sheet. Clearly and neatly shade the correct option.

1. An engineer drives a test vehicle over the same distance at two different speeds: 60 km/h on the first segment and 40 km/h on the second segment. The average speed over the entire trip is not the arithmetic mean of 60 and 40. Instead, it is given by the harmonic mean as:
 - a. 48 km/h
 - b. 40 km/h
 - c. 38 km/h
 - d. 28 km/h
2. Two manufacturing plants produce the same metal component with an average length of 50 mm. Plant A has a standard deviation of 0.2 mm, while Plant B has a standard deviation of 1.5 mm. Which plant produces more reliable components?
 - a. Plant A, because its measurements vary less around the mean
 - b. Plant B, because it has a larger standard deviation
 - c. Both plants are equally reliable because their averages are equal
 - d. Reliability cannot be determined from standard deviation
3. You are part of a team assessing the impact of industrial wastewater on a nearby river. The river stretches over 50 kilometres and passes through residential, agricultural, and industrial zones. Which sampling technique is most appropriate for accurately assessing water quality along the entire stretch of the river?
 - a. Simple Random Sampling
 - b. Convenience Sampling
 - c. Stratified Sampling
 - d. Snowball Sampling

4. An engineer concludes a machine is working perfectly because the average output meets specifications. Why might this conclusion be misleading?
- a. The machine may still have high variability
 - b. The sample size is always irrelevant
 - c. The mean always represents the entire population perfectly
 - d. The standard deviation cannot be calculated
5. Why is variability important in engineering production?
- a. It helps reduce the number of measurements needed
 - b. It determines the consistency and reliability of the process
 - c. It replaces the need for averages
 - d. It eliminates the need for sampling
6. Statistical thinking helps engineers primarily to:
- a. Remove uncertainty completely
 - b. Make decisions using data despite uncertainty
 - c. Avoid collecting data
 - d. Replace engineering judgement
7. A fair coin is tossed three times. What is the probability of obtaining at least one head?
(*Hint: Use the complement probability of no heads.*)
- a. $\frac{1}{8}$
 - b. $\frac{3}{8}$
 - c. $\frac{7}{8}$
 - d. $\frac{1}{2}$
8. Two fair dice are rolled. Which of the following events is more likely?
- a. The sum is 7
 - b. The sum is 2
 - c. The sum is 12
 - d. All events are equally likely
9. A couple has two children. You are told that one of the children is a boy. What is the probability that the other child is a girl?

a. $\frac{1}{2}$

c. $\frac{2}{3}$

b. $\frac{1}{3}$

d. $\frac{3}{4}$

10. A card is drawn randomly from a standard deck of 52 cards. Given that the card drawn is red, what is the probability that it is a heart?

a. $\frac{1}{13}$

c. $\frac{1}{2}$

b. $\frac{1}{4}$

d. $\frac{3}{4}$

11. Find the number of ways in which a committee of 4 can be chosen from 6 boys and 5 girls if it must consist of 2 boys and 2 girls.

a. 150

c. 120

b. 100

d. 200

12. Which of the following is **not** a random variable?

a. Number of defective components in a batch

c. The probability that a machine fails

b. The weight of a metal sample

d. The time required to complete a machining process

13. A discrete random variable X has the following probabilities:

X	0	1	2
$P(X)$	0.2	0.5	0.4

Which statement is correct?

a. This is a valid probability distribution

c. This is invalid because probabilities sum to 1.1

b. This is invalid because probabilities exceed 1

d. This is invalid because probabilities cannot be decimals

14. A game pays \$10 if a head appears when a fair coin is tossed, and \$0 if a tail appears. What is the expected payoff?
- a. \$10 b. \$5 c. \$0 d. \$2.50
15. A remote village has 1000 vehicles with sequential registration numbers starting from 1000. Out of the total, 30% are without pollution clearance certificates. Even- and odd-numbered vehicles are operated on even- and odd-numbered dates respectively. If 100 vehicles are chosen at random on an even-numbered date, the expected number without a pollution clearance certificate is:
- a. 15 c. 30
b. 10 d. 50
16. Which of the following variables is most likely to follow a normal distribution?
- a. Number of defective bolts in a box c. Heights of adult males in a population
b. Number of cars passing a checkpoint per minute d. Number of children in a family
17. During an experiment, 80% of tractor engines were damaged in the first trial. The percentage of engines surviving any later experiment is half that of the preceding one. Find the probability that an engine will survive four experiments.
- a. 0.08 b. 0.05 c. 0.45 d. 0.11
18. The hardness of a certain alloy (measured on the Rockwell scale) is a random variable X . Assume that $X \sim U(50, 70)$. Find $E(X)$.
- a. 75.2 b. 65.2 c. 62.5 d. 60.0
19. For the function $f(x) = a + bx$, $0 \leq x \leq 1$, to be a valid probability density function, which of the following is correct?
- a. $a = 1, b = 4$ b. $a = 0.5, b = 1$ c. $a = 0, b = 1$ d. $a = 1, b = -1$
20. A drawer contains 3 white socks and 3 black socks. If 2 socks are picked at random without replacement, what is the chance of getting a non-matching pair?
- a. 0.3 b. 0.6 c. 0.7 d. 0.9

21. Which of the following violates the binomial assumptions?
- | | |
|------------------------------------|--|
| a. Tossing a coin 5 times | c. Selecting students without replacement from a small class |
| b. Inspecting 20 bulbs for defects | d. Testing 15 batteries for failure |
22. A coin was flipped 1000 times, and 550 times it showed heads. What is the most reasonable conclusion about whether the coin is biased?
- | | |
|---|--|
| a. The coin is definitely biased because heads occurred more than tails | c. The coin may or may not be biased because random variation can produce unequal outcomes |
| b. The coin is definitely fair because 1000 is a large number of trials | d. The coin must always produce exactly 500 heads and 500 tails if it is fair |
23. A factory produces bolts where each bolt is independently defective with probability 0.05. In a batch of 20 bolts, what is the probability that **exactly 2** are defective?
- | | |
|---------------------------------------|--------------------------|
| a. $\binom{20}{2}(0.05)^2(0.95)^{18}$ | c. $(0.05)^2(0.95)^{18}$ |
| b. $\binom{20}{2}(0.05)^{18}(0.95)^2$ | d. $20 \times (0.05)^2$ |
24. The number of equipment failures in a plant follows a Poisson distribution with mean $\lambda = 3$ per month. What is the probability of no failures in a given month?
- | | |
|--------------|-----------------|
| a. e^{-3} | c. $1 - e^{-3}$ |
| b. $3e^{-3}$ | d. e^3 |
25. A machine produces cables with breaking strength $X \sim N(500, 25^2)$ N. A cable is **rejected** if its strength falls below 450 N. What is the probability a cable is **accepted**?
- | | |
|-------------------------------|------------------------------|
| a. $P(Z < -2) \approx 0.0228$ | c. $P(Z < 2) \approx 0.9772$ |
| b. $P(Z > 2) \approx 0.0228$ | d. $P(Z > -2) \approx 0.50$ |
26. Which of the following is **not** a condition required for a Binomial distribution?
- | | |
|--|---|
| a. Fixed number of trials n | c. Trials are independent |
| b. Probability of success varies per trial | d. Each trial results in success or failure |

27. A continuous random variable X has pdf $f(x) = 3x^2$ for $0 \leq x \leq 1$. What is $E(X)$?

a. $\frac{1}{2}$

c. $\frac{2}{3}$

b. $\frac{3}{4}$

d. $\frac{1}{4}$

28. For a random variable X with $E(X) = 5$ and $E(X^2) = 30$, what is $\text{Var}(X)$?

a. 25

c. 5

b. 30

d. 6

29. Let X be the number of trials until the first success in independent Bernoulli trials with $p = 0.4$. What is $E(X)$?

a. 0.4

c. 4

b. 2.5

d. 1.6

30. Two random variables X and Y are **independent**. Which of the following must be true?

a. $E(XY) = E(X) + E(Y)$

c. $E(XY) = E(X) \cdot E(Y)$

b. $\text{Var}(X + Y) = \text{Var}(X) \cdot \text{Var}(Y)$

d. $\text{Cov}(X, Y) > 0$

31. An estimator $\hat{\theta}$ is said to be **unbiased** if:

a. $\hat{\theta} = \theta$ always

c. $E(\hat{\theta}) = \theta$

b. $\text{Var}(\hat{\theta}) = 0$

d. $\hat{\theta}$ has the smallest variance among all estimators

32. A 95% confidence interval for a population mean is constructed from a sample. Which interpretation is **correct**?

a. There is a 95% probability the true mean lies in this specific interval

c. If many samples were taken, 95% of such intervals would contain the true mean

b. 95% of the data lies within this interval

d. The true mean equals the sample mean with 95% accuracy

33. As sample size n increases, the width of a confidence interval (at a fixed confidence level):
- a. Increases, because more data adds uncertainty
 - b. Stays the same
 - c. Decreases, because $\sigma/\sqrt{n}$ gets smaller
 - d. Depends only on the confidence level, not n
34. A 90% confidence interval gives [42, 58] and a 99% CI from the *same* data gives [38, 62]. Which statement is **correct**?
- a. The 99% interval is more precise and therefore preferred
 - b. The 90% interval is incorrect because it is narrower
 - c. Higher confidence requires a wider interval, trading precision for certainty
 - d. Both intervals must have the same width for valid inference
35. A researcher tests $H_0 : \mu = 50$ vs. $H_a : \mu \neq 50$ and obtains $p\text{-value} = 0.03$. At $\alpha = 0.05$, the correct conclusion is:
- a. Fail to reject H_0 ; there is insufficient evidence
 - b. Accept H_0 ; the mean is 50
 - c. Reject H_0 ; there is sufficient evidence that $\mu \neq 50$
 - d. Accept H_a ; the mean is definitely not 50
36. In hypothesis testing, a Type II error occurs when:
- a. H_0 is rejected when it is true
 - b. H_0 is rejected when it is false
 - c. H_0 is not rejected when it is false
 - d. The p -value exceeds α and H_0 is true
37. An engineer tests whether a new alloy exceeds a standard mean of 400 MPa using $n = 25$ samples with **unknown** σ . Which test is most appropriate?
- a. Two-tailed z -test
 - b. Two-tailed t -test with 25 degrees of freedom
 - c. One-tailed t -test with 24 degrees of freedom
 - d. Chi-squared test
38. The **power** of a hypothesis test is defined as:

- a. $P(\text{Reject } H_0 \mid H_0 \text{ is true}) = \alpha$ c. $P(\text{Reject } H_0 \mid H_0 \text{ is false}) = 1 - \beta$
- b. $P(\text{Accept } H_0 \mid H_0 \text{ is false}) = \beta$ d. The probability that the sample mean equals μ_0
39. In simple linear regression $\hat{y} = b_0 + b_1x$, the slope b_1 represents:
- a. The value of y when $x = 0$ c. The average change in y for a one-unit increase in x
- b. The correlation between x and y d. The variance of the residuals
40. A regression analysis yields $R^2 = 0.85$. The most accurate interpretation is:
- a. 85% of the data points lie on the regression line c. 85% of the variability in y is explained by x
- b. The slope of the regression line is 0.85 d. There is an 85% probability the model is correct

SECTION B

Carefully Read Each Question and Attempt Only ONE (1) Question

Question 1

- (a) Consider an exploration company that is determined to discover two new oil fields in a virgin basin it is prospecting and will drill as many exploratory holes as required until two discoveries are made. Assume that the probability of success (discovery) for each exploratory hole is 0.1, and that the outcomes of the holes are independent.
- i. What type of probability distribution is appropriate for modeling this situation? Explain briefly why this distribution is suitable
 - ii. Define the random variable X appropriate for this problem.
 - iii. Find the probability that exactly two holes are required to obtain the two discoveries.
 - iv. Find the probability that the company will require exactly three holes to obtain the two discoveries.
 - v. Interpret what the result in part (iv) means in the context of exploratory drilling.
- (b) Show that the Binomial distribution has $Variance = p(1 - p)$.

Question 2

- (a) An environmental monitoring agency measures the concentration of dissolved oxygen (DO) in a river system as part of a water quality assessment. The specifications require that the mean DO level must be 8.0 mg/L to sustain aquatic life. From historical data, the standard deviation of DO measurements is known to be 0.8 mg/L. A geomatics engineer collects a random sample of 25 water specimens from designated monitoring stations along the river and obtains a sample mean DO level of 7.4 mg/L. A significance level of $\alpha = 0.05$ is used.
- Clearly state the null and alternative hypotheses for this test, identifying whether it is a one-tailed or two-tailed test.
 - Compute the test statistic and identify the critical region at $\alpha = 0.05$.
 - What conclusions should be drawn regarding the water quality of the river? Justify your answer statistically.
 - Suppose the true mean DO level is actually 7.6 mg/L. What type of error, if any, might the original test have committed? Explain the practical implication of this error in the context of environmental management.

Question 3

- (a) A construction company bids on projects. The probability of winning a bid is 0.4. Each successful bid earns a profit of \$80,000 while each unsuccessful bid costs \$5,000 in preparation fees. If the company submits 10 bids:
- Find the expected profit from a single bid
 - Find the expected total profit from all 10 bids
 - Find the variance of profit from a single bid
- (b) A structural engineer measures the tensile strength of steel cables. From 40 samples, the mean breaking strength is 850 kN with a standard deviation of 25 kN. Construct a 99% confidence interval for the true mean breaking strength. If the cable is rated safe up to 800 kN. What does your interval say about safety?

— END OF EXAMINATION —

Descriptive Statistics	
Mean (Average)	$\mu = \frac{\sum X_i}{n}$
Variance	$\sigma^2 = \frac{\sum (X_i - \mu)^2}{n}$
Standard Deviation	$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{n}}$
Probability Rules	
Addition Rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication Rule	$P(A \cap B) = P(A) P(B A) = P(B) \cdot P(A B)$
Bayes' Theorem	$P(A B) = \frac{P(B A) P(A)}{P(B)}$
Discrete Distributions	
Binomial	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Poisson	$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$
Negative Binomial	$P(X = k) = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$
Bernoulli	$P(X = k) = p^k (1 - p)^{1-k}$
Geometric	$P(X = k) = (1 - p)^{k-1} \cdot p \quad P(X \leq k) = 1 - (1 - p)^k$
Continuous Distributions	
Normal	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
Uniform	$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b$
Standard Normal (Z-score)	$Z = \frac{X - \mu}{\sigma}$
Confidence Intervals & Hypothesis Testing	
CI for μ (σ known)	$CI = \bar{x} \pm z \cdot \frac{\sigma}{\sqrt{n}}$
CI for μ (σ unknown)	$CI = \bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$
Test Statistic for Mean	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$